Day - 20

Percentage Based on MARKS

1) In an examination, there were 1000 boys and 800 girls. 60% of the boys and 50% of the girls passed. Find the percent of the candidates failed?

(boys) 1000 ⇓ 60%

girls (800) ⇓ 50%

failed ⇒ 40% ⇓ 400 + 50% ⇓ 400 ⇒ 800

Total boys and Girls ⇒ 1800

$$\frac{\cancel{800}^{4}}{\cancel{1800}_{9}} \times 100 \Rightarrow \frac{400}{9}$$

$$\Rightarrow 44.\overline{44}$$

2) In an examination, a candidate must secure 40% marks to pass. A candidate, who gets 220marks, fails by 20 marks. What are the max marks for exam.

Passed mark => 240

40% => 240

100% => x

6

$\cancel{4}x = \cancel{24}0 \times 10$

$\boxed{x = 600}$

3) For an exam, it is required to get 36% of max marks to pass. A student got 113 marks and failed by 85 marks. The max marks for the examination are:

113 + 85

passed mark => 198

36% = 198

100% = x

$\cancel{18}$ 22

$\cancel{36}x = \cancel{198} \times 100$

= 550

4) A student has to obtain 33% of total marks to pass. He got 25% of total marks failed by 40 marks. The number total marks is:

$$25\% \rightarrow 8\% \leftarrow 33\%$$

$$\downarrow$$

$$40$$

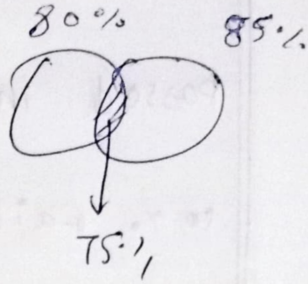
$$\cancel{8}\% = \cancel{40}^{5}$$

$$100\% = x$$

$$x = 500$$

5) In exam, 80% of the boys passed in English and 85% passed in math, while 75% passed in both. If 45 boys failed in both. The number of boys who sat for the exam was

$= 80+85-75$



$= 90\%$

Boys passed 90%

$1\cancel{0}\% = 45$

$10\cancel{0}\% = x$

$x = 450$

6) In exam, 65% of the student passed in maths. 48% passed in physics and 30% passed in both. How much percent of student failed in both subjects?

$65+48-30 \Rightarrow 83\%$

$\Downarrow$

difference

17% fail

7) In an exam, 70% of the candidates passed in English. 80% passed in maths. 10% failed in both the subject. If 144 candidates passed in both, the total number of (c) were:

70%	80%	%	10
Eng	M	↓ 144	fail

$100\% = 70\% + 80\% + 10\% - x\%$

$100\% = 160\% - x\%$

$\boxed{x = 60\%}$

$60\% = 144$ (24)

$100\% = x$

$x = 240$